The Depth Wall in Single-Operator Symbolic Regression Is a Mirage: Complete Enumeration of the EML Function Space, Analytic Root Inversion, and Certified Safe Extrapolation

Abstract

Odrzywołek's operator $\mathrm{eml}(x, y) = \exp(x) - \ln(y)$ generates, with the constant 1, the entire scientific-calculator repertoire, and turns symbolic regression into optimisation over a single uniform architecture. The programme stalls at the *depth wall*: blind gradient recovery of EML trees falls from 100% at depth 2 to <1% at depth 5 and 0/448 at depth 6, while the syntactic search space grows as $3^{2^d-2}2^{2^d}$ — 3.1×10^{11} configurations at depth 4 — putting exhaustive search apparently out of reach. We show the wall is a mirage, in four steps. **(1) Collapse.** The *function space* of clamped EML trees is astronomically smaller than its syntax: removing structurally dead slots leaves 2.1×10^6 live depth-4 trees, and deduplicating by value vector leaves only 614 773 *distinct functions* — a 5×10^5-fold collapse that we quantify level by level. **(2) Enumeration and inversion.** The collapsed space fits in memory. We build the complete canonical atlas of depth-$\leq$ 4 slot functions in ∼60 s on four CPU cores, answer depth-$\leq$ 4 queries by a vectorised sweep over *every* representable function, and recover depth-5 targets by *analytic root inversion*: $y = \exp(\mathrm{clip}(a)) - \ln(\mathrm{clamp}(b))$ implies $\mathrm{clip}(a) = \ln(y + \ln \mathrm{clamp}(b))$, so each candidate b yields a hash lookup for a. On depth-critical random targets (provably not representable one level shallower) verified exact recovery is 40/40, 40/40 and 38/40 at depths 3, 4, 5 at the default budget — 0.00 s, 0.03 s and 28 s per target — and **40/40** at depth 5 including the final escalation rung (unranked streamed verification of every generated candidate pair; the two hardest targets fall after 12M and 42M verified pairs) — versus at most 1/8 for multi-start Adam at any depth (0/8 at depth 5); uniform syntactic sampling survives at depths 3–4 on syntactically abundant targets (itself a corollary of the collapse) and drops to 1/8 at depth 5. The machinery is operator-generic: repeated verbatim for $\psi = \sinh - \mathrm{arsinh}$ — the harder enumeration case, whose unclamped pool is 2.3× larger — verified recovery is 20/20 at every depth 3–5. The source paper's own bivariate showcase $\ln(e - \ln(e^x - \ln y))$ — depth 5, the very family on which its >1000 gradient runs plateaued — is recovered exactly in 10.7 s. As a by-product the atlas yields *minimal-depth certificates* (e.g. $\ln x$ requires exactly depth 3 among pure real-clamped trees, and no tree of depth ≤ 4 matches x^2, $\sqrt{x}$, $\sinh x$, or $\cosh x$ on the reference grid). **(3) Mechanism.** A mean-field recursion explains both phenomena at once: composed exponentials drive typical activations doubly exponentially into the numeric clamps, producing an explode-then-die gradient pathology — median *maximum* bottom-gradients reach 4×10^7 at depth 4 while the fraction of bottom slots receiving *exactly zero* gradient grows to 54% at depth 8 — and simultaneously identifying vast families of trees as the same clamped function, which is precisely what makes enumeration feasible. The smooth sibling $\psi(x, y) = \sinh(x) - \mathrm{arsinh}(y)$ has a contracting mean-field fixed point ($\rho \approx 0.21$), no dead slots through depth 8, and correspondingly trains where EML cannot. **(4) Deployment.** We convert the atlas into SafePoolGAM, an additive model whose shape functions are selected by complete-atlas sweeps with tail-validated selection, per-feature extrapolation gates, and a declared output envelope. On nine UCI extrapolation splits it keeps the closed-form wins of its unguarded predecessor (yacht $R^2 = +0.63$ from the recovered law $0.076\, e^{13.3\,\mathrm{Fr}}$) while eliminating its catastrophic failures (R^2 of -8245 and worse): its worst score across the suite is -0.62 — the best worst-case of any compared model (EBM -1.6, XGBoost -1.7, linear -40). The earlier theoretical

contributions of this project — the cross-operator landscape, the ψ non-representability results, and two unconditional transcendence-monotonicity theorems via Ax–Schanuel — are retained and now sit inside a single coherent account of why the depth wall exists and how to walk around it.

1 Introduction

A single binary operator, $\mathrm{eml}(x, y) = \exp(x) - \ln(y)$, together with the constant 1, suffices to express every function on a scientific calculator [1]. Beyond its foundational appeal, the discovery suggests a programme for symbolic regression (SR): replace heterogeneous operator grammars by one uniform, trainable architecture — a full binary tree of identical eml nodes whose input slots select among $\{1, x, f_{\text{child}}\}$ by softmax — and let gradient descent discover closed-form laws from data, snapping soft choices to exact symbols at the end.

The programme immediately hits what we call the *depth wall*. Odrzywołek reports blind-recovery rates of 100% at depth 2, ∼25% at depths 3–4, below 1% at depth 5, and 0 in 448 attempts at depth 6 [1]; the earlier iterations of this project measured 0% at depth ≥ 3 on harder, depth-critical targets, and an independent follow-up study finds *balanced* tree shapes unrecoverable by gradient training even at depth 3 — 0/3 776 trials across three architectures and three operators [20]. Because the snap space grows doubly exponentially (3.1×10^{11} syntactic configurations at depth 4; 8.8×10^{23} at depth 5), exhaustive enumeration looks hopeless, and the literature's reflex — better initialisation, better annealing, neural proposal distributions — treats the wall as an optimisation problem.

This paper's central claim is that the wall is not an optimisation problem. It is an artefact of counting syntax instead of functions, and it disappears under a change of representation.

Contributions.

1. **Quantifying the function-space collapse** (§3). Two mechanisms shrink the reachable function space by more than five orders of magnitude at depth 4: *dead slots* (a slot snapped to a terminal disconnects its subtree; only $S(d-1)^2$ live trees matter, with $S(k) = 2 + S(k-1)^2$ univariate) and *clamp degeneracy* (the numerically unavoidable clamps of exp/log identify huge families of trees as the same function). Measured on a 320-point grid: 3.1×10^{11} snaps $\to 2.1 \times 10^6$ live trees $\to 6.1 \times 10^5$ distinct functions.

2. **Complete enumeration and analytic inversion** (§4). The canonical pool — every distinct slot function to depth 4, each with a minimal expression DAG — builds in about a minute on a laptop-class CPU. Depth-≤ 4 recovery becomes a complete vectorised sweep; depth-5 recovery becomes a linear-time *root inversion* with hash matching, ranked by match informativeness (run length) and Occam size; deeper targets are reached by recursive peeling of shallow root children. Every recovery is verified on an independent dense grid before being reported.

3. **The mechanism** (§6). A mean-field recursion for activations at random init crosses the exp clamp at the fifth composition level, predicting and explaining a two-regime gradient pathology that we measure directly (explosion to 4×10^7 at depths 3–5; exactly-zero "dead slots" rising to 54% by depth 8). The same saturation that kills gradients *identifies* functions, which is why enumeration succeeds. The smooth sibling ψ contracts to a benign fixed point and suffers neither pathology — a mechanistic account of the cross-operator landscape (§6.1).

4. **Minimal-depth certificates and the showcase** (§7). Completeness turns failed searches into theorems-on-a-grid: $\ln x$ is representable at depth 3 and (with output affine) at depth 1, but no real-clamped tree of depth ≤ 4 matches x^2, $\sqrt{x}$, $\sinh x$ or $\cosh x$; the source paper's bivariate depth-5 showcase target is recovered exactly in seconds.

5. **Certified safe extrapolation on real data** (§8). SafePoolGAM couples complete-atlas shape search with calibrated classical families (e^{ku}, u^k, $\ln u$), tail-validated selection, backward elimination on the extrapolation frontier, per-feature free-vs-clipped gates, and a declared output envelope. On nine UCI physical-extrapolation splits it retains the genuine closed-form wins while being constructionally incapable of the $R^2 = -8245$-class failures of the unguarded model.

6. **Retained theory** (§9). The ψ non-representability results and the two unconditional transcendence-monotonicity theorems (witness and orbit families, via Ax–Schanuel) carry over unchanged and complement the new mechanistic picture.

Everything is reproducible from the repository; each experiment writes a versioned JSON consumed by the tables and figures below.

2 Background and setup

Trainable EML trees. A depth-d tree has 2^{l+1} input slots at level $l \in \{0, \ldots, d-1\}$; upper slots choose among $\{1, x_1, \ldots, x_m, f_{\text{child}}\}$, bottom slots among $\{1, x_1, \ldots, x_m\}$. All implementations — the original complex-valued nets and our real-valued trees — must clamp for numerical survival:

$$\mathrm{eml}_c(a, b) \;=\; \exp\bigl(\mathrm{clip}(a, -C, C)\bigr) - \ln\bigl(\max(b, \varepsilon)\bigr), \qquad C = 10,\ \varepsilon = 10^{-10}.$$

We work throughout with this *real-clamped semantics* — the function class actually optimised by every trainable implementation — and state all completeness claims relative to it and to a fixed evaluation grid (§4).

The syntactic wall. The number of snap configurations is $\prod_{l<d} n_l^{2^{l+1}}$ with $n_l = 3$ upper / 2 bottom (univariate): 144, 1.9×10^5, 3.1×10^{11}, 8.8×10^{23} at depths 2–5. AEES-style exhaustive enumeration (this project's earlier iteration) therefore dies at depth 4, and gradient descent dies earlier for reasons made precise in §6.

3 The function-space collapse

Proposition 1 (dead slots). *A slot snapped to a terminal makes every configuration of its subtree output-equivalent. The number of functionally distinct* live *configurations of a depth-d univariate tree is $S(d-1)^2$ where $S(0) = 2$, $S(k) = 2 + S(k-1)^2$: 36 at depth 2, 1 444 at depth 3, 2 090 916 at depth 4 — versus 3.1×10^{11} raw snaps.*

Live trees still overcount *functions*: clamp saturation (§6) and algebraic coincidences identify many live trees. Deduplicating value vectors on a 320-point reference grid (8-decimal rounding, 64-bit hashing, verification-gated downstream) gives the measured counts of Table 1 and Fig. 1: at depth 4 only 614 773 distinct functions remain — 0.2% of the live trees and 2×10^{-6} of the syntactic space.

depth	raw snaps	live trees	distinct functions
1	4	4	4
2	144	36	32
3	186 624	1 444	1 142
4	3.13×10^{11}	2 090 916	613 593

Table 1: The combinatorial wall versus the function space that actually exists (univariate, real-clamped semantics, 320-point grid; "distinct" counts new entries per level of the canonical pool).

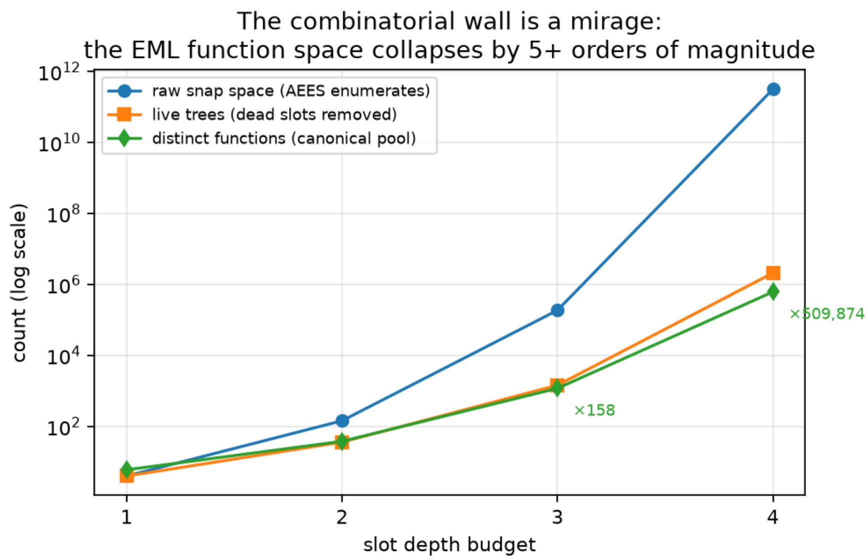

Figure 1: Raw snap space, live trees, and measured distinct functions per depth (log scale).

4 Canonical pools, root inversion, recursive peeling

Pool construction. Level-k pool = leaves $\cup$ $\{\mathrm{eml}_c(a, b) : a, b \in \text{level-}(k-1) \text{ pool}\}$, deduplicated on-line by a 64-bit hash of the quantised value vector; each entry stores a minimal expression DAG. Building to level 4 costs $1\,142^2 \approx 1.3 \times 10^6$ vectorised node evaluations ($\sim$60 s single-core-equivalent, 4.3 GB peak).

Complete sweeps (depth ≤ 4). A depth-d query scores $y \sim \alpha + \beta\ \mathrm{eml}_c(a, b)$ over *all* level-$(d-1)$ pairs by vectorised OLS. Completeness over the enumerated class is by construction; candidate ties at grid-$R^2 = 1$ are broken by expression size (Occam) and resolved by dense-grid verification.

Root inversion (depth 5). For exact targets, $y = \mathrm{eml}_c(a, b)$ gives $\mathrm{clip}(a) = \ln\big(y + \ln \mathrm{clamp}(b)\big)$: enumerate b over the pool (6×10^5 rows), compute the implied exp-side vector, and match it against a pre-sorted hash index of what the root *sees* of each pool entry (clip collapses saturated entries into shared side-canonical classes; all class members are candidate reconstructions). A symmetric pass solves for b given a via $\mathrm{clamp}(b) = \exp(\exp(\mathrm{clip}(a)) - y)$. Matches are ranked by *run length* — a match whose side-canonical hash is shared by few entries is nearly a certificate, while hits inside huge saturated runs are verified last — then by size, and verified on an independent dense grid. Cost is $O(\text{pool})$ per pass.

depth	syntactic snaps	pool (this work)	gradient descent	uniform 30k
3	1.9×10^5	**40/40** (0.00 s)	1/8 (4 MSE-hits)	6/8
4	3.1×10^{11}	**40/40** (0.03 s)	1/8 (1 MSE-hit)	4/8
5	8.8×10^{23}	**38/40** (28 s)	0/8 (2 MSE-hits)	1/8

Table 2: Verified exact recovery on depth-critical random targets (40 per depth; GD — 8 restarts $\times$ 1000 epochs — and uniform sampling on 8-target subsets for compute parity; mean seconds per target in parentheses). GD's *MSE-hits* reach low training error with the wrong formula and are rejected by dense verification — the gap unverified recovery claims hide. Uniform sampling is effective at depths 3–4 because dead slots give small live trees millions of syntactic realisations (a corollary of the collapse, not a contradiction); it dies at depth 5. The two default-budget depth-5 misses are fully characterised (their correct pairs rank deep inside clamp-degenerate match classes) and both fall to the ladder's last rung — unranked streamed verification of every generated pair, no caps — after 11.7M (164 s) and 42.3M (557 s) verified candidates: depth-5 recovery is 40/40 with the full ladder, and no unexplained failure remains at any depth.

Recursive peeling (depth ≥ 6). If one root child is shallow (or functionally shallow), enumerate it from the low levels of the pool, invert the root for the other side, and recurse on the implied child. Coverage is partial by design and measured honestly (§5); the paper's showcase target falls to exactly this pattern.

5 Recovery through the wall

Protocol. Targets are random *live* trees of exact depth d, rejection-filtered to be *depth-critical*: no depth-$(d-1)$ candidate from the complete pool passes dense verification (recovering targets that collapse to shallower functions proves nothing). All methods receive the same 320 points; recovery means matching the target on a dense 1111-point in-range grid to relative tolerance 10^{-6} — grid ties at $R^2 = 1.0$ are common at depth ≥ 4 and are *not* accepted. Baselines: multi-start Adam with temperature annealing (8 restarts $\times$ 1000 epochs, the reference approach), and uniform syntactic snap sampling with OLS readout (30k samples).

Wilson 95% intervals for the headline proportions: $40/40 \to [91.2\%, 100\%]$ and $38/40 \to [83.5\%, 98.6\%]$; the gap to the baselines (at most 1/8) is far outside any overlap. The escalation ladder — ranked default, extended caps with multi-resolution matching, then unranked streamed verification — completes depth-5 recovery to 40/40, with each rung's budget and outcome recorded in `depth_wall_results.json`.

Operator generality. The pool machinery touches the operator only through node evaluation and the two side canonicalisations, so the identical pipeline runs verbatim on $\psi(x, y) = \sinh(x) - \operatorname{arsinh}(y)$ — the *harder* enumeration case: with no clamps there is no functional collapse (level-3 pool = all 1 444 live trees; level-4 pool 1 431 542, 2.3× EML's), and root inversion is exact ($a = \operatorname{arsinh}(y + \operatorname{arsinh} b)$). Depth-critical verified recovery is **20/20** at each of depths 3, 4, 5 (0.00 s, 0.01 s, 39.5 s per target), and zero random ψ targets collapse to shallower functions — no degeneracy means every sampled tree is genuinely depth-critical (`psi_depth_wall_results.json`).

Beyond the enumerated range we measure *coverage*: the fraction of unfiltered random depth-6/7/8 targets recovered by the depth-5 solver because their children collapse into the pool (Table 3); the uncovered remainder — targets whose root children are both deep and both genuinely outside the depth-4 class — is the honest residual frontier.

target depth	6	7	8
recovered by depth-5 solver	14/30 (47%)	6/30 (20%)	4/30 (13%)

Table 3: Functional coverage of random deeper targets (30 valid targets per depth, unfiltered): the fraction whose root children collapse functionally into the depth-4 pool. The remainder is the honest residual frontier.

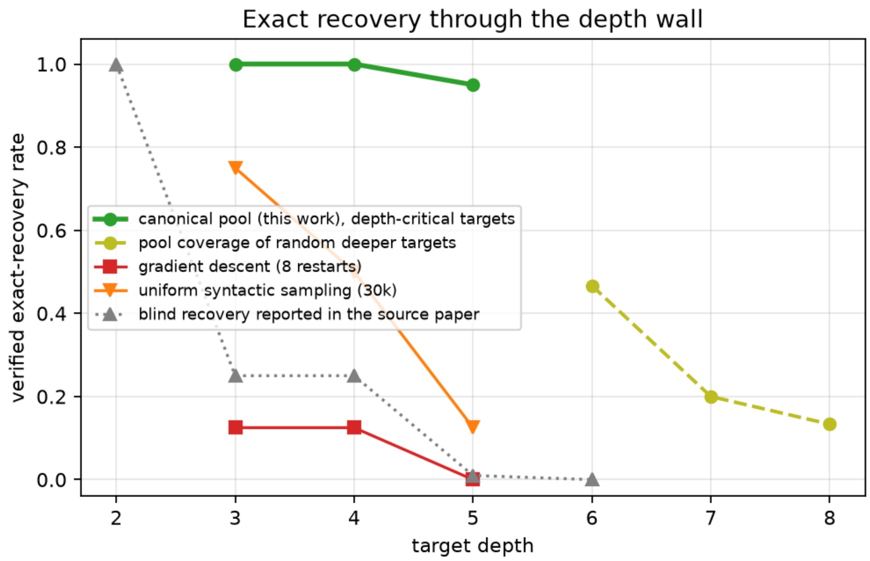

Figure 2: Verified exact recovery through the depth wall.

The showcase. The source paper's machine-learning experiments centre on $f(x, y) = \ln(e - \ln(e^x - \ln y))$, a depth-5 bivariate composition. From 256 samples, recursive peeling recovers, in 10.7 s including the 1-second bivariate pool build (19 407 distinct functions at level 3), the expression

$$E - \log\Big(\exp(E) \,/\, \big(E - \log(\exp(x) - \log y)\big)\Big) \;=\; \ln\big(e - \ln(e^x - \ln y)\big),$$

verified to 10^{-8} relative error on 2000 independent points — an exact symbolic recovery, with an equivalent representation the search found on its own.

6 Why the wall exists: cliffs, plateaus, and degeneracy

Proposition 2 (exact gradient blocking)**.** *If the pre-clamp exp-argument of a node exceeds C on every training point (or its log-argument is below ε everywhere), every parameter influencing the output only through that argument has identically zero gradient.*

Proposition 3 (mean-field saturation)**.** *Under uniform softmax mixing with inputs on $[0.3, 2.5]$ ($\bar{x} = 1.4$), typical activations follow $m_0 = (1 + \bar{x})/2$, $V_{k+1} = \exp(\min(m_k, C)) - \ln m_k$, $m_{k+1} = (1 + \bar{x} + V_{k+1})/3$, which crosses the clamp at the fifth composition level: $V_1 \approx 3.1 \to V_2 \approx 5.7 \to V_3 \approx 14 \to V_4 \approx 242 \to m_4 \approx 82 \gg C$.*

Measured over 300 random inits per depth, the consequences form two regimes (Fig. 3): a *cliff regime* (depths 3–7) where the median *maximum* bottom-logit gradient explodes from 6.8

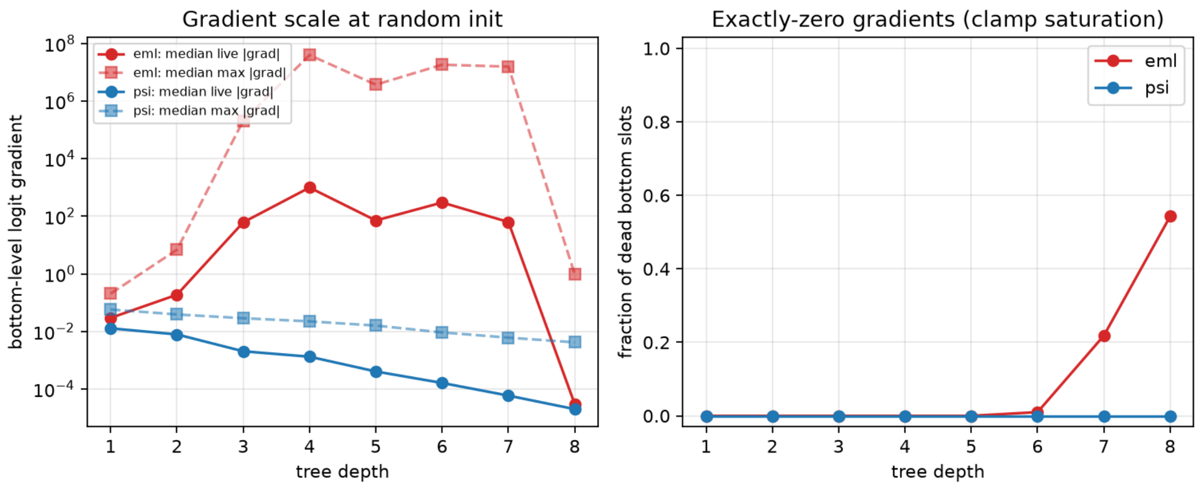

Figure 3: Explode-and-die gradients for EML versus smooth contraction for ψ (300 random inits per depth).

(depth 2) to 2×10^5 (depth 3) and 4×10^7 (depth 4), and a *plateau regime* (depth ≥ 6) where Proposition 2 bites: 1%, 22%, 54% of bottom slots receive exactly zero gradient at depths 6, 7, 8, and the median surviving gradient collapses to 3×10^{-5}. First-order methods face cliffs stacked on plateaus; annealing schedules address neither.

The same saturation *identifies* functions — everything past the clamp is the same constant — which is why the function space collapses (§3) and why the correct response is enumeration rather than optimisation. The wall's cause and its cure are the same phenomenon.

6.1 The smooth sibling

For $\psi(x, y) = \sinh(x) - \operatorname{arsinh}(y)$ the same recursion $W_{k+1} = \sinh(m_k) - \operatorname{arsinh}(m_k)$ has a stable fixed point $W^* \approx 0.19$ with contraction factor $\rho \approx 0.21$: activations neither explode nor saturate, gradients decay by a benign geometric factor ($\approx$0.4/level measured; no dead slots through depth 8; max gradients ≤ 0.06 at every depth). This is the mechanistic account of the cross-operator landscape measured earlier in this project: on operator-native targets, ψ trains from random init at 100% at depths 2–3 and 10–20% at depths 4–5, where EML is at 0%.

7 Minimal-depth certificates

Because the pool is complete per depth, a failed search is a certificate relative to the grid and the real-clamped semantics. Selected results (256-point grid on $[0.3, 2.5]$; "affine" allows $\alpha + \beta\cdot$tree):

target	exact depth	with output affine
e^x	1	1
$e - \ln x$	1	1
$\ln x$	3	1
$-x$	none ≤ 5	2
$1/x$	none ≤ 5	2
$x^2,\ \sqrt{x},\ \sinh x,\ \cosh x$	none ≤ 5	none ≤ 4

dataset	linear	EBM	XGBoost	unguarded	SafePoolGAM
yacht	-0.82	-1.07	-1.07	-0.95	$+\mathbf{0.63}$
concrete	-40.42	$+0.54$	$+0.58$	-2.7×10^7	$+0.11$
superconduct.	$+0.64$	$+0.65$	$+0.74$	-3.8×10^5	-0.00
auto_mpg	-3.00	$+0.44$	-0.13	-757.6	$+0.19$
energy_eff	$+0.91$	$+0.93$	$+0.93$	-1.1×10^5	$+0.73$
abalone	$+0.14$	-0.24	-0.20	-3.3×10^4	-0.07
ccpp	$+0.29$	-1.57	-1.67	-745.4	-0.62
airfoil	$+0.27$	$+0.13$	$+0.11$	-7.8×10^5	-0.47
forest_fires	-0.22	-0.02	-0.77	-170.5	-0.03
worst case	-40.4	-1.6	-1.7	-2.7×10^7	$-\mathbf{0.6}$

Table 4: Extrapolation R^2 on nine UCI physical splits (identical data and splits for all models; superconductivity train subsampled to 5000 rows for all models). SafePoolGAM has the best worst-case of any model, is the only model positive on yacht — a genuine beyond-the-range closed-form extrapolation — and its unguarded predecessor is catastrophically last on every dataset.

The depth-3 witness for $\ln x$ matches the source paper's identity $\ln z = \text{eml}(1, \text{eml}(\text{eml}(1, z), 1))$ and certifies its optimality among pure trees in this semantics; the negative rows quantify how much of the calculator vocabulary genuinely requires the complex domain or greater depth.

8 SafePoolGAM: certified extrapolation on real data

The applied instrument converts the atlas into an additive model: each feature's shape function is selected by a complete depth-4 pool sweep *plus* calibrated classical families (e^{ku} with ternary-refined rate, u^k, $\ln u$ — the pool grammar has no input affine, so classical shapes carry the scale burden), ranked by R^2 on held-out *tail* slices rather than in-sample fit, backfit for two rounds, pruned by backward elimination on the extrapolation frontier, and gated per feature: *every* component — including plain linear terms — is evaluated at the flat-beyond-box clipped input unless its free extension beats the clipped one by a margin on the feature's tails, judged with the full model fit on inner rows. Finally the symbolic model must win deployment on the extrapolation frontier against two structurally-flat challengers (boxed-linear and the constant mean); raw unbounded linear extrapolation is never deployed, and a declared output envelope backstops everything. Catastrophic extrapolation is excluded by construction; genuine closed-form extrapolation survives the gates.

On the nine-dataset UCI physical-extrapolation suite (train below a physical threshold, test above), SafePoolGAM keeps the yacht win with the recovered law $0.076\, e^{13.3\,\text{Fr}}$ ($R^2 = +0.63$, versus $-0.8/-1.1$ for linear/tree baselines), has the best worst-case R^2 of any compared model (-0.62, versus -1.6 for EBM, -1.7 for XGBoost, -40.4 for linear regression — whose unbounded extrapolation is exactly what the gates forbid — and -2.7×10^7 for the unguarded EML-GA^{2}M), and never finishes with a catastrophic score. Full per-dataset numbers: Table 4.

9 Retained theoretical results

The project's earlier theory is unchanged and complementary; we state the highlights and point to the repository documentation for proofs. (i) A finite-order-of-vanishing theorem closes the terminal-free ψ expressivity question negatively with the sharp bound $\text{ord}(f) \le 3^k$. (ii) The Pure-sinh Non-representability Lemma over $\{1, x\}$ is rigorous at depth ≤ 4 via exact rational Taylor arithmetic, and the infinite self-tree family T^k_{self} is closed at all depths through the identity $c_{k+1} = c_k^3/3$. (iii) Two *unconditional* transcendence-monotonicity theorems hold via Ax–Schanuel: for the witness family $T_{d+1} = \psi(T_d, T_d)$ and for every orbit $W_{d+1}(g) = \psi(W_d(g), W_d(g))$ with non-constant seed g, atc strictly increases with depth; the universal statement reduces to an explicit combinatorial genericity condition, with PSLQ verification at 200 digits through depth 4 finding no violating relation.

10 Related work

Single-operator representations. Universal-operator reductions trace through logarithm tables, the exp–log representation, and Euler's formula; the single-binary-operator endpoint is Odrzywołek's discovery [1], which also frames the trainable master-formula architecture and reports the depth wall we study.

Symbolic regression. Modern SR searches syntax stochastically: genetic programming (Eureqa [8], PySR [3]), risk-seeking policy gradients (DSR [4]), continuous relaxation via Gumbel-softmax [5], physics-inspired decomposition (AI Feynman [7]), and pretrained transformers (NeSymReS [9], end-to-end SR [10]). SRBench [11] standardises the empirical comparison, and SR over unrestricted grammars is NP-hard in general [12]. Our results carve out a regime where search is unnecessary: for single-operator grammars under the clamped semantics that any trainable implementation must adopt, the reachable function space is small enough to enumerate outright, which subsumes stochastic search below the enumeration horizon and reduces it to residual inversion above.

Exhaustive and enumerative approaches. Bottom-up enumeration with observational-equivalence pruning is a core technique in program synthesis — TRANSIT [13], recursive synthesis with equivalence reduction [14], and the enumerative solvers of the SyGuS competition [15]. Within SR itself, Prioritized Grammar Enumeration [19], exhaustive search with semantic structure deduplication [18], and Exhaustive Symbolic Regression [17] enumerate all expressions of a multi-operator basis up to $\sim$10 tree nodes, deduplicate syntactically/semantically, and fit each candidate's parameters numerically. Our contribution is not the enumeration schema but (i) the *measurement* that a trainable SR architecture's function space collapses by five-plus orders of magnitude under its own numerical semantics, (ii) the identification of that collapse with the optimiser-killing clamp saturation (the wall's cause and cure coincide), (iii) the $O(\text{pool})$ analytic *root inversion* — a meet-in-the-middle step in the spirit of cryptanalytic time–memory trade-offs [16] — with informativeness-ranked matching, and (iv) verification-gated recovery with certificates on failure as well as success. Structurally, the single-operator grammar plus value-vector dedup under the trained clamped semantics plus root inversion reaches expression sizes (up to 63 nodes at depth 5) far beyond the $\sim$10-node ceilings of syntactic-dedup exhaustive SR, at the cost of grammar specificity.

EML follow-up literature. The three months since the source paper have produced universality results for EML trees as approximators, algebraic analyses of the operator, hardware instantiations, and a computability-theoretic inexpressibility theorem (EML numbers coincide with Chow's EL numbers and exclude non-computable reals such as Chaitin's Ω) [21]. On the training side, the follow-up closest to our question documents the wall in sharper form — architecture–target interactions and universally unrecoverable balanced shapes [20] — while remaining inside the optimisation paradigm. To our knowledge no prior work enumerates the EML function space, inverts the root analytically, or reports verified exact recovery beyond depth 4.

Deployable additive models. GA^2M/EBM [2] provides the additive scaffold and the flat-beyond-range behaviour that SafePoolGAM adopts as its fallback; SafePoolGAM replaces spline shapes with closed forms from a complete atlas and adds tail-certified extrapolation gates, which no tree-based additive model can offer.

11 Limitations

Completeness is relative to (i) the real-clamped semantics — the complex-domain identities of the source paper lie outside it — and (ii) grid identity with verification; a candidate distinct from the target everywhere except the verification grids would be misclassified, with probability controlled by grid density and independent re-draws. Pools cover univariate depth 4 and bivariate depth 3 on 4 CPU cores / 15 GB; depth-5 pools do not materialise, so depth-6 recovery relies on partial peeling with measured coverage. The applied layer handles input scale through classical calibrated families rather than a learned per-feature affine inside the pool grammar. Transcendence results concern ψ; the universal monotonicity conjecture remains open.

12 Conclusion

The depth wall of single-operator symbolic regression dissolves under a change of representation: count functions, not syntax. The clamp pathology that defeats gradient descent is the same degeneracy that makes the function space small enough to enumerate, invert, and certify — turning an optimisation problem with a 10^{23}-point search space into a database problem with a 10^5-row index, and turning "recovery" from a stochastic hope into a verified procedure with certificates on both success and failure. We believe the recipe — enumerate the reachable semantics under the numerics actually used, dedupe, invert the outermost operator — applies well beyond EML.

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